

## LECTURE 15 (OCTOBER 27TH)

**Definition.** (Time-dependent perturbation theory) In time-dependent perturbation theory, the Hamiltonian depends on time as

$$\hat{H}(t) = \hat{H}^{(0)} + \lambda \hat{H}^{(1)}(t)$$

we assume that we know all information about  $\hat{H}^{(0)}$ , with  $\{E_n^{(0)}, |\phi_n^{(0)}\rangle\}$ . We can imagine like a laser being shot into a hydrogen atom and transitions occurring, similar to scattering and collisions. The goal is to of course solve

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$$

with  $|\psi(t)\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} |\phi_n\rangle$ . Substituting, we have a coupled ordinary differential equation with respect to  $c_n(t)$ . Our first goal will be to derive this. We have

$$\begin{aligned} i\hbar \sum_n \left( \dot{c}_n(t) + \left( -\frac{iE_n}{\hbar} \right) c_n(t) \right) e^{iE_n t/\hbar} |\phi_n\rangle \\ = \sum_n \left( E_n^{(0)} + \lambda \hat{H}^{(1)}(t) \right) c_n(t) e^{-iE_n t/\hbar} |\phi_n\rangle \end{aligned}$$

Applying  $\langle \phi_m |$  on both sides,

$$i\hbar \dot{c}_m(t) e^{-iE_m t/\hbar} = \sum_n \lambda \langle \phi_m | \hat{H}^{(1)}(t) | \phi_n \rangle e^{-iE_n t/\hbar} c_n(t)$$

We conclude

$$i\hbar \frac{dc_m(t)}{dt} = \sum_n e^{i(E_m - E_n)t/\hbar} \lambda H_{mn}^{(1)}(t) c_n(t)$$

which is exact. In reality, in many cases, at  $t = t_i$ , only one state is occupied, say  $|\phi_k\rangle$ , and

$$c_n(t_i) = \delta_{nk}$$

Let's now approximate starting here.

$$i\hbar \frac{dc_{m \neq k}(t)}{dt} \approx e^{i(E_m - E_k)t/\hbar} H_{mk}^{(1)}(t) c_k(t)$$

is the first order approximation as we expect the initial energy to be accumulated at  $E_k$ . Solving,

$$c_{m \neq k}(t) - c_{m \neq k}(t_i) = \int_{t_i}^t dt' \frac{dc_{m \neq k}(t')}{dt'} = \int_{t_i}^t dt' \frac{1}{i\hbar} e^{i(E_m - E_k)t'/\hbar} V_{mk}(t')$$

Let's return to the original ansatz. We want the transition amplitude  $\langle \phi | \psi(t) \rangle = c_m(t) e^{-iE_m t / \hbar}$ . Where the transition probability is given by  $|\langle \phi_m | \psi(t) \rangle|^2 = |c_m(t)|^2$ . Notice how

$$\sum_m |c_m(t)|^2 = \sum_m \langle \psi(t) | \phi_m \rangle \langle \phi_m | \psi(t) \rangle = \langle \psi(t) | \psi(t) \rangle = 1$$

and the coefficients are normalised.

**Example.** (Harmonic oscillator) Consider the simple harmonic oscillator with  $\lambda H^{(1)}(t') = qEx e^{-(t')^2/\tau^2}$  where  $\tau$  can be interpreted as pulse width. Take  $t_i = -\infty$  and the ground state  $n = 0$ . Up to the first order, we have

$$c_{n \neq 0}(t) = \frac{qE}{i\hbar} \int_{-\infty}^t dt' e^{i\omega_{n0}t'} \langle n | x | 0 \rangle e^{-(t')^2/\tau^2}$$

where  $\omega_{mn} = (E_m - E_n)/\hbar$ . Also, notice that  $\langle n | x | 0 \rangle$  is not zero only if  $n = 1$ . Computing the Gaussian integral,

$$P_m(\infty) = \pi \left( \frac{qE\tau}{\hbar} \right)^2 \times |\langle m | x | 0 \rangle|^2 \times e^{-\omega_{k0}^2 \tau^2 / 4}$$

which is only nonzero for  $m = 1$ .

**Example.** (Other example) Let

$$\hat{H}^{(1)}(t) = \left( \hat{M} e^{i\omega t} + \hat{M}^\dagger e^{i\omega t} \right)$$

where we assume that  $\hat{M} = \hat{M}^\dagger$  for simplicity. Then,

$$\hat{H}^{(1)}(t) = \hat{M} (e^{-i\omega t} + e^{i\omega t})$$

For first order, let  $t_i = 0$  and  $c_n(0) = \delta_{nk}$ . We then have

$$\begin{aligned} c_{m \neq k}(t) &= \int_0^t dt' \frac{1}{i\hbar} e^{i\omega_{mk}t'} \langle m | M | k \rangle \left( e^{-i\omega t'} + e^{i\omega t'} \right) \\ &= \frac{1}{i\hbar} M_{mk} \frac{e^{i(\omega_{mk} \mp \omega)t} - 1}{i(\omega_{mk} \mp \omega)} \\ &= \frac{1}{i\hbar} M_{mk} \frac{1}{i(\omega_{mk} \mp \omega)} e^{i(\omega_{mk} \mp \omega)t/2} \left( e^{i(\omega_{mk} \mp \omega)t/2} - e^{-i(\omega_{mk} \mp \omega)t/2} \right) \\ &= \frac{1}{i\hbar} M_{mk} \left( \frac{\sin((\omega_{mk} \mp \omega)t/2)}{(\omega_{mk} \mp \omega)t/2} \right)^2 e^{i(\omega_{mk} \pm \omega)t/2} \end{aligned}$$

## LECTURE 16 (OCTOBER 29TH)

**Theorem.** (Fermi Golden Rule) Finishing last time's calculations,

$$|c_{m \leftarrow k}(t)|^2 = \frac{1}{\hbar^2} \left[ \frac{2 - 2 \cos((\omega_{mk} - \omega)t)}{(\omega_{mk} - \omega)^2} |M_{mk}|^2 + \frac{2 - 2 \cos((\omega_{mk} + \omega)t)}{(\omega_{mk} + \omega)^2} |M_{mk}^\dagger|^2 \right]$$

as  $t \rightarrow \infty$ . We can write the above in terms of sine squared and we have

$$|c_{m \leftarrow k}|^2 = \frac{1}{\hbar^2} \left[ \frac{\sin^2((\omega_{mk} - \omega)t/2)}{((\omega_{mk} - \omega)/2)^2} + \dots \right]$$

where

$$\lim_{t \rightarrow \infty} \frac{\sin^2((\omega_{mk} - \omega)t/2)}{((\omega_{mk} - \omega)/2)^2} = 2\pi t \delta(\omega_{mk} - \omega)$$

Now,

$$|c_{m \leftarrow k}(t)|^2 = \frac{2\pi t}{\hbar^2} \left[ \delta(\omega_{mk} - \omega) |M_{mk}|^2 + \delta(\omega_{mk} + \omega) |M_{mk}^\dagger|^2 \right]$$

recalling that  $\omega_{mk} = (E_m - E_k)/\hbar$ , we can write

$$|c_{m \leftarrow k}(t)|^2 = \frac{2\pi t}{\hbar} \left[ \delta(E_m - E_k - \hbar\omega) |M_{mk}|^2 + \delta(E_m - E_k + \hbar\omega) |M_{mk}^\dagger|^2 \right]$$

which gives the transition probability per time, the transition rate, as

$$\Gamma = \left( \frac{2\pi}{\hbar} \right) \left[ \delta(E_m - E_k - \hbar\omega) |M_{mk}|^2 + \delta(E_m - E_k + \hbar\omega) |M_{mk}^\dagger|^2 \right]$$

which is called the Fermi Golden Rule. Let's further investigate this and check units. We want it to be  $[\Gamma] = T$ . We see that, indeed,

$$[\Gamma] = \frac{1}{E \cdot T} \times \frac{1}{E} \times E^2 = \frac{1}{T}$$

In the general case,

$$\Gamma(i \rightarrow f) = \frac{2\pi}{\hbar} \sum \delta(\Sigma E) |M|^2$$

where the sum is over all final states.

**Example.** (Two-state harmonic perturbation) For simplicity, let  $V_{11} = V_{22} = 0$  for some perturbing Hamiltonian  $\hat{V}$ . Let

$$V_{12} = \gamma e^{i\omega t}$$

and let  $\gamma > 0$ . Then,  $V_{21} = \gamma^* e^{-i\omega t}$  where  $\gamma^* = \gamma$ . Up to first order, the time-dependent perturbation becomes

$$|\psi(t)\rangle = \sum_{n=1,2} c_n(t) e^{-iE_n t/\hbar} |\phi_n\rangle$$

The differential equation we need to solve is:

$$\begin{cases} i\hbar \frac{dc_1}{dt}(t) = V_{12}e^{i\omega_{12}t}c_2(t) \\ i\hbar \frac{dc_2}{dt}(t) = V_{21}e^{i\omega_{21}t}c_1(t) \end{cases}$$

where  $\omega_{12} = -\omega_{21}$ . Setting  $\omega - \omega_{21} = \Omega$ , we can write

$$\begin{aligned} i\hbar \frac{dc_1}{dt}(t) &= \gamma e^{i\Omega t} c_2(t) \\ i\hbar \frac{dc_2}{dt}(t) &= \gamma e^{-i\Omega t} c_1(t) \end{aligned}$$

Solving the coupled differential equations,

$$i\hbar \ddot{c}_2(t) = r \left( -i\Omega e^{-i\Omega t} \frac{i\hbar \dot{c}_2(t)}{\gamma} e^{i\Omega t} + e^{-i\Omega t} \frac{\gamma}{i\hbar} e^{i\Omega t} c_2(t) \right)$$

we obtain

$$\ddot{c}_2(t) = -i\Omega \dot{c}_2(t) - \frac{\gamma^2}{\hbar^2} c_2(t)$$

The general solution can be obtained through substituting  $c_2(t) = Ae^{i\alpha t}$ . We get the equation  $-\alpha^2 = \alpha\Omega - \gamma^2/\hbar^2$  giving two real solutions for  $\alpha$ .

$$\alpha_{\pm} = \frac{-\Omega \pm \Delta}{2}$$

where  $\Delta = \sqrt{\Omega^2 + 5\gamma^2/\hbar^2}$ . With the initial conditions  $c_1(0) = 1$  and  $c_2(0) = 0$ ,

$$c_2(t) = c_+ e^{i\alpha_+ t} + c_- e^{-i\alpha_- t}$$

with  $c_+ = -c_-$  and

$$c_2(t) = c_+ \left( e^{i\alpha_+ t} - e^{i\alpha_- t} \right) = (2i)c_+ e^{i(\alpha_+ + \alpha_-)t/2} \sin \left( \frac{\Delta t}{2} \right)$$

where the condition  $c_1(0) = 1$  gives  $c_+ = -\gamma/\Delta\hbar$ . Thus,

$$|c_2(t)|^2 = \frac{\gamma^2/\hbar^2}{\frac{\gamma^2}{\hbar^2} + \left( \frac{\omega - \omega_{21}}{2} \right)^2} \sin^2 \left( \sqrt{\frac{\gamma^2}{\hbar^2} + \left( \frac{\omega - \omega_{21}}{2} \right)^2} t \right)$$

which is known as the Rabi formula.

**Definition.** (Interaction representation) We see a middle ground representation of the Heisenberg and Schrodinger picture. The idea is simple – we write  $\hat{H} = \hat{H}_0 + \lambda \hat{V}(t)$  and

only let the time evolve via  $\lambda\hat{V}(t)$ . We express

$$|\alpha, t_0, t\rangle_I = e^{i\hat{H}_0 t/\hbar} |\alpha, t_0, t\rangle_S$$

and

$$|\alpha, t_0, t_0\rangle_S = e^{-iE_\alpha t_0/\hbar} |\alpha\rangle$$

which is important to remember. Then,

$$\hat{V}_I(t) = e^{iH_0 t/\hbar} V_S e^{-iH_0 t/\hbar}$$

## LECTURE 17 (NOVEMBER 3RD)

**Definition.** (Interaction picture) Let  $\hat{H} = \hat{H}_0 + \lambda V(t)$ . Let

$$|\alpha, t_0, t\rangle_I = e^{iH_0 t/\hbar} |\alpha, t_0, t\rangle_S$$

and

$$|\alpha, t_0, t_0\rangle_S = e^{-iE_\alpha t_0/\hbar} |\alpha\rangle$$

as the definitions of a interaction picture. We then have

$$i\hbar \frac{\partial}{\partial t} |\alpha, t_0, t\rangle_I = V_I(t) |\alpha, t_0, t\rangle_I$$

and

$$\frac{dV_I}{dt} = \frac{1}{i\hbar} [V_I, H_0] + e^{iH_0 t/\hbar} \frac{\partial V_S}{\partial t} e^{-iH_0 t/\hbar}$$

where  $(H_0)_I = H_0$ .

*Proof.*

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\alpha, t_0, t\rangle_I &= \left( i\hbar \frac{\partial}{\partial t} e^{iH_0 t/\hbar} \right) |\alpha, t_0, t\rangle_S + e^{iH_0 t/\hbar} \left( i\hbar \frac{\partial}{\partial t} |\alpha, t_0, t\rangle_S \right) \\ &= -H_0 e^{iH_0 t/\hbar} |\alpha, t_0, t\rangle_S + e^{iH_0 t/\hbar} (H_0 + V(t)) |\alpha, t_0, t\rangle_S \\ &= e^{iH_0 t/\hbar} V(t) |\alpha, t_0, t\rangle_S = e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar} e^{iH_0 t/\hbar} |\alpha, t_0, t\rangle_S \\ &= V_I(t) |\alpha, t_0, t\rangle_I \end{aligned}$$

□

**Corollary.** Previously, we have seen how

$$|\psi(t)\rangle_S = \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle$$

Then,

$$\begin{aligned} |\psi(t)\rangle &= e^{iH_0 t/\hbar} |\psi(t)\rangle_S = e^{iH_0 t/\hbar} \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle \\ &= \sum_n c_n(t) |n\rangle \end{aligned}$$

**Corollary.** In quantum field theory, the time evolution operator  $U_I(t, t_0)$  resulted in

$$|\alpha, t_0, t\rangle_I = U_I(t, t_0) |\alpha, t_0, t_0\rangle_I$$

applying this to the time evolution equation for the interaction picture ket from the above,

$$i\hbar \frac{\partial}{\partial t} U_I(t, t_0) = V_I(t) U_I(t, t_0)$$

applying the perturbation here, we obtain the Dyson series or the Dyson expansion.

$$U_I^{(0)}(t, t_0) = \mathbf{1}$$

and

$$U_I(t, t_0) = \mathbf{1} + \left( -\frac{i}{\hbar} \right) \int_{t_0}^t dt' V_I(t') U_I(t', t_0)$$

from the differential equation gives

$$U_I^{(1)} = \mathbf{1} + \left( -\frac{i}{\hbar} \right) \int_{t_0}^t dt' V_I(t')$$

up to the first order. The second order is trivial as

$$\begin{aligned} U_I^{(2)} &= \mathbf{1} + \left( -\frac{i}{\hbar} \right) \int_{t_0}^t dt' V_I(t') U_I^{(1)}(t') \\ &= \mathbf{1} + \left( -\frac{i}{\hbar} \right) \int_{t_0}^t dt' V_I(t') + \left( -\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt' V_I(t') \int_{t_0}^{t'} dt'' V_I(t'') \end{aligned}$$

**Definition.** (Berry phase) Berry phase is closely relevant to topology in physics. The geometry of quantum states is given by  $|\Psi(R(t))\rangle \sim e^{i\theta} |\Psi(R(t))\rangle$  where  $R(t)$  is a set of parameters (such as  $\mathbf{R}$  for a line and  $S^2$  for perhaps applied field direction). Imagine that we slowly vary the parameters, thus looking at an adiabatic transition. That is, we take  $\Delta > \hbar\omega$  where  $\Delta$  is the difference of energy levels. For instantaneous eigenkets, we have

$$H(R) |n(R)\rangle = E_n(R) |n(R)\rangle$$

## LECTURE 18 (NOVEMBER 10TH)

**Definition.** We have previously defined

$$F_{\mu\nu}^{(n)} = i \left[ \sum_{\text{all}, n} \langle \frac{\partial n}{\partial R^\mu} | n' \rangle \langle n' | \frac{\partial n}{\partial R^\nu} \rangle - \sum_{\text{all}, n} \langle \frac{\partial n}{\partial R^\nu} | n' \rangle \langle n' | \frac{\partial n}{\partial R^\mu} \rangle \right]$$

where

$$\langle n | \frac{\partial H}{\partial R} | n' \rangle = (E_n - E_{n'}) \langle \frac{\partial n}{\partial R} | n' \rangle$$

for  $n \neq n'$ . What about the case where  $n' = n$ ?

$$\langle \frac{\partial n}{\partial R^\mu} | n \rangle \langle n | \frac{\partial n}{\partial R^\nu} \rangle - \langle \frac{\partial n}{\partial R^\nu} | n \rangle \langle n | \frac{\partial n}{\partial R^\mu} \rangle = (ia)(ib) - (-ib)(-ia) = 0$$

Thus, we can change the summation index above from  $n' \neq n$ . We obtain

$$F_{\mu\nu}^{(n)} = i \sum_{n' \neq n} \left[ \frac{\langle n | \frac{\partial H}{\partial R^\mu} | n' \rangle \langle n' | \frac{\partial H}{\partial R^\nu} | n \rangle}{(E_n - E_{n'})^2} - (\dots)^* \right]$$

This is the most direct formula. Summing both sides over with respect to  $n$ , we can then change  $n$  with  $n'$  and find that the summation over  $n$  is zero. for convenience, let's define a  $2 \times 2$  Herimian matrix that is traceless iwth basis  $O_x, O_y, O_z$  as the basis.

$$H = h_x O_x + h_y O_y + h_z O_z = \begin{pmatrix} h_z & h_x - ih_y \\ h_x + ih_y & -h_z \end{pmatrix}$$

We take  $\vec{h} = (h_x, h_y, h_z) \in \mathbf{R}$  to be real and we slowly vary the parameters. The eigenvalues are  $-h$  and  $h$  where  $h = \sqrt{h_x^2 + h_y^2 + h_z^2}$ . Let's now express everything in spherical coordinates  $h_x = \sin \theta \cos \phi$ ,  $h_y = h \sin \theta \sin \phi$ , and  $h_z = h \cos \theta$ . Consider the eigenstates  $|-\rangle$  and  $|+\rangle$ . Recall that there is phase ambiguity in making the choices

$$|-\rangle = \begin{pmatrix} \sin(\theta/2)e^{-i\phi} \\ -\cos(\theta/2) \end{pmatrix}$$

and

$$|+\rangle = \begin{pmatrix} \cos(\theta/2)e^{-i\phi} \\ \sin(\theta/2) \end{pmatrix}$$

The nontrivialness of the parameter space's topology leads to non-uniqueness of global eigenvectors!

**Corollary.** (In the test) Let's calculate the Berry phase. We can easily check that  $\langle \pm | \vec{O} | \pm \rangle = \pm \vec{h}/h$ . For example,  $\langle - | O_z | - \rangle = \sum^2(\theta/2) - \cos^2(\theta/2) = -\cos^2 \theta = -h_z/h$ . Notice that

$$A_\theta^{(-)} = \langle - | i \frac{\partial}{\partial \theta} | - \rangle = 0$$

and

$$A_{\phi}^{(-)} = \langle - | i \frac{\partial}{\partial \phi} | - \rangle = \sin^2(\theta/2) = \frac{1}{2}(1 - \cos \theta)$$

We will in the far future learn that  $1/2$  actually denotes spin. Recall that the field strength was defined as

$$F_{\theta\phi}^{(-)} = \partial_{\theta} A_{\phi} - \partial_{\phi} A_{\theta} = \frac{1}{2} \sin \theta$$

In the language of differential forms,

$$F^{(-)} = \frac{1}{2} \sin \theta d\theta \wedge d\phi$$

which we can rewrite as

$$F^{(-)} \frac{S}{h^2} (h^2 \sin \theta d\theta \wedge d\phi)$$

Integrating over the sphere,

$$\int_{S^2} F^{(-)} d\vec{S} = \frac{s}{h^2} (4\pi h^2) = 4\pi s$$

We find  $\int \vec{B} \cdot d\vec{S} = \int_{S^2} F^{(-)}$  and

$$\vec{B} = \frac{s}{h^2} \hat{h}$$

where the magnetic sitting at  $h = 0$  is unique. By the way, check  $F^{(+)} = -F^{(-)}$ .

**Example.** We try to obtain the above by this time using the direct formula. Recall that  $F_{xy} = B_z$ . Using

$$F_{\mu\nu}^{(-,+)} = i \sum_{m \neq n} \left( \left\langle n \left| \frac{\partial H}{\partial R^{\mu}} \right| m \right\rangle \left\langle m \left| \frac{\partial H}{\partial R^{\nu}} \right| n \right\rangle - (\dots)^* \right)$$

and

$$\frac{\partial H}{\partial h_x} = O_x \quad \& \quad \frac{\partial H}{\partial h_y} = O_y$$

At the end of the day,

$$F_{xy} = \frac{1}{2} \frac{h_z}{h^3}$$

which I want you to check!

**Corollary.** Let  $H = \vec{p}^2/2m + V(\vec{x}, t)$ . What is  $\vec{A}$ ? Recall that  $m\vec{a} = \vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ . We find that

$$H = \frac{(\vec{p} - q\vec{A})^2}{2m} + q\phi$$

which would be memorised.